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\\Research

Investigators - validated on FIT

Frederick

High bandwidth Siemens product scans

localizer tse_MDME_p3 tof_cs_acc12.0_large angio_twist_sag_iso 4d_flow_aorta_retro_pre 4d_flow_aorta_retro_post ep2d_tra_hemo t1_space_sag_cs4_iso B1Map_for_T1_mapping T1Map_anatomical T1Map_p2_anatomical T2Map_anatomical T2StarMap_anatomical T2StarMap_p2_anatomical
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\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\localizer

TA: 19 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	7

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1

Inline - Subtraction

StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	40 deg
Measurements	1
Contrasts	1
TE	5.00 ms
TR	20.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\tse_MDM
E_p3

TA: 4:25 min Coil Selection: Auto Voxel Size: 0.7×0.7×4.0 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	4350.0 ms
TE 1	23.00 ms
TE 2	101.00 ms
Averages	1
AutoAlign	---

Contrast - Common

TR	4350.0 ms
TE 1	23.00 ms
TE 2	101.00 ms
Magn. Preparation	Slice-sel. IR
Flip Angle	150 deg
Fat-Water Contrast	Standard
Contrasts	2
Wrap-up Magn.	None
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	4
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s

Contrast - Dynamic

Multiple Series	Off
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Resolution - Common

FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
Base Resolution	320
Phase Resolution	80 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	TSE/Separate
Acceleration Factor PE	3
Reference Lines PE	32

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	30
Distance Factor	25 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	230 mm
FOV Phase	81.3 %
Slice Thickness	4.0 mm
TR	4350.0 ms
Multi-Slice Mode	Interleaved

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Subtraction

Subtract	Off
Measurements	4
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	MDME
Dimension	2D
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	150 Hz/Px
Echo Spacing	11.26 ms
Define	Turbo Factor
Turbo Factor	5

Sequence - Part 2

Phase Correction	Automatic
Reduce Motion Sens.	Off

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\tof_cs_ac
c12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm

Geometry - Common

FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\angio_tw
ist_sag_iso

TA: 1:32 min Coil Selection: Auto Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 6 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	On
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2.26 ms
TE	0.92 ms
AutoAlign	---

Contrast - Common

TR	2.26 ms
TE	0.92 ms
Flip Angle	25 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	TWIST
Central Region A	15 %
Sampling Density B	20 %
Reconstruction Scheme	Symmetric Share
Temporal Resolution	1.99 s
Temporal Interpolation	1
Measurements	40
Pause after Meas. 1	0.0 s
Pause after Meas. 2	0.0 s
Pause after Meas. 3	0.0 s

Contrast - Dynamic

Pause after Meas. 4	0.0 s
Pause after Meas. 5	0.0 s
Pause after Meas. 6	0.0 s
Pause after Meas. 7	0.0 s
Pause after Meas. 8	0.0 s
Pause after Meas. 9	0.0 s
Pause after Meas. 10	0.0 s
Pause after Meas. 11	0.0 s
Pause after Meas. 12	0.0 s
Pause after Meas. 13	0.0 s
Pause after Meas. 14	0.0 s
Pause after Meas. 15	0.0 s
Pause after Meas. 16	0.0 s
Pause after Meas. 17	0.0 s
Pause after Meas. 18	0.0 s
Pause after Meas. 19	0.0 s
Pause after Meas. 20	0.0 s
Pause after Meas. 21	0.0 s
Pause after Meas. 22	0.0 s
Pause after Meas. 23	0.0 s
Pause after Meas. 24	0.0 s
Pause after Meas. 25	0.0 s
Pause after Meas. 26	0.0 s
Pause after Meas. 27	0.0 s
Pause after Meas. 28	0.0 s
Pause after Meas. 29	0.0 s
Pause after Meas. 30	0.0 s
Pause after Meas. 31	0.0 s
Pause after Meas. 32	0.0 s
Pause after Meas. 33	0.0 s
Pause after Meas. 34	0.0 s
Pause after Meas. 35	0.0 s
Pause after Meas. 36	0.0 s
Pause after Meas. 37	0.0 s
Pause after Meas. 38	0.0 s
Pause after Meas. 39	0.0 s
Multiple Series	Off
Time to Center	8.5 s
Burn Time to Center	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	70 %
Slice Resolution	75 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	3
Reference Lines 3D	24
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Strong
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	160
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	2.26 ms

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Maximum
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Subtraction

Subtract	On
Subtraction Mode	Standard
Save Subtracted Images	On
Subtrahend	1
Subtraction Group	1
Measurements	40
Subtraction Indices	
Autoscaling	Off
Scaling Factor	3
Offset	0
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
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Inline - MIP

MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fldyn
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Bandwidth	750 Hz/Px
Asymmetric Echo	Strong
Optimization	Min. TE TR

Sequence - Part 2

RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\4d_flow_
aorta_retro_pre

TA: 19 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-8 %
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
TR	40.56 ms
TE	2.73 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	40.56 ms
TE	2.73 ms
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	5 deg
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
Base Resolution	160
Phase Resolution	85 %
Slice Resolution	86 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	39
Acceleration Factor 3D	1
Reference Lines 3D	18
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Strong
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-8 %
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
TR	40.56 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L3.0 A30.0 H5.4
L	3.0 mm
A	30.0 mm
H	5.4 mm
Initial Orientation	S > C
S > C	26.30
> T	3.20
Initial Rotation	0.00 deg

Geometry - Navigator

Navigator	1
Position	R75.0 P41.3 H6.3 mm
Orientation	Sagittal
Rotation	0.00 deg
Base Size Phase	400 mm
Base Size Read	400 mm
Thickness	10 mm
Navigator	2
Position	R75.0 P41.3 H6.3 mm
Orientation	Sagittal
Rotation	0.00 deg
Base Size Phase	400 mm
Base Size Read	400 mm
Thickness	10 mm

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	5 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm

System - Adjustments

Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	ECG/Retro
Average Cycle	No Signal ms
Calculated Phases	25
TR	40.56 ms
Segments	2
Concatenations	1
Arrhythmia Detection	None

Physio - Cardiac

Magn. Preparation	None
Dark Blood	Off
FOV Read	400 mm
FOV Phase	82.5 %
Phase Resolution	85 %
Cine	On
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Gate
Scout Mode	On
Scout Duration	19 s
Scout TR	100 ms
Accept Window ±	6.00 mm
Accept Position (green)	128.00 mm
Search Window ±	40.00 mm
Search Position (red)	118.00 mm
Tracking Factor	0.60
Chronological Position	Before Echo

Physio - PACE

RF Pulse Type	Crossed Pair
Resp. Motion Adaptation	On
Concatenations	1

Flow

Flow Mode	Single Vel.
Encodings	3
Velocity Enc.	150 cm/s
Direction 1	Through Plane
Direction 2	A >> P
Direction 3	F >> H
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	Off
Phase Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	Slab-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Reordering	Linear
Bandwidth	496 Hz/Px
Echo Spacing	5.07 ms
Asymmetric Echo	Strong
Optimization	Min. TE TR
Define	Segments
Segments	2

Sequence - Part 2

RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE TR

Sequence - Part 2

Introduction	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\4d_flow_
aorta_retro_post

TA: 19 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: 3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	On
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	-8 %
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
TR	40.56 ms
TE	2.73 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	40.56 ms
TE	2.73 ms
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	15 deg
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Contrast - Dynamic

Reordering	Linear
------------	--------

Resolution - Common

FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
Base Resolution	160
Phase Resolution	85 %
Slice Resolution	86 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	39
Acceleration Factor 3D	1
Reference Lines 3D	18
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Strong
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	-8 %
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	16.7 %
FOV Read	400 mm
FOV Phase	82.5 %
Slice Thickness	2.50 mm
TR	40.56 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L3.0 A30.0 H5.4 mm
Orientation	S > C26.3 > T3.2
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	L3.0 A30.0 H5.4
L	3.0 mm
A	30.0 mm
H	5.4 mm
Initial Orientation	S > C
S > C	26.30
> T	3.20
Initial Rotation	0.00 deg

Geometry - Navigator

Navigator	1
Position	R75.0 P41.3 H6.3 mm
Orientation	Sagittal
Rotation	0.00 deg
Base Size Phase	400 mm
Base Size Read	400 mm
Thickness	10 mm
Navigator	2
Position	R75.0 P41.3 H6.3 mm
Orientation	Sagittal
Rotation	0.00 deg
Base Size Phase	400 mm
Base Size Read	400 mm
Thickness	10 mm

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	5 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm

System - Adjustments

Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	ECG/Retro
Average Cycle	No Signal ms
Calculated Phases	25
TR	40.56 ms
Segments	2
Concatenations	1
Arrhythmia Detection	None

Physio - Cardiac

Magn. Preparation	None
Dark Blood	Off
FOV Read	400 mm
FOV Phase	82.5 %
Phase Resolution	85 %
Cine	On
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Gate
Scout Mode	On
Scout Duration	19 s
Scout TR	100 ms
Accept Window ±	6.00 mm
Accept Position (green)	128.00 mm
Search Window ±	40.00 mm
Search Position (red)	118.00 mm
Tracking Factor	0.60
Chronological Position	Before Echo

Physio - PACE

RF Pulse Type	Crossed Pair
Resp. Motion Adaptation	On
Concatenations	1

Flow

Flow Mode	Single Vel.
Encodings	3
Velocity Enc.	150 cm/s
Direction 1	Through Plane
Direction 2	A >> P
Direction 3	F >> H
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	Off
Phase Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	Slab-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Reordering	Linear
Bandwidth	496 Hz/Px
Echo Spacing	5.07 ms
Asymmetric Echo	Strong
Optimization	Min. TE TR
Define	Segments
Segments	2

Sequence - Part 2

RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE TR

Sequence - Part 2

Introduction	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\lep2d_tra_hemo

TA: 6 sec Coil Selection: Auto Voxel Size: 0.9x0.9x5.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	6120.0 ms
TE	30.00 ms
Averages	1
Concatenations	1
AutoAlign	Head > Brain

Contrast - Common

TR	6120.0 ms
TE	30.00 ms
MTC	Off
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	5.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	6120.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.01 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.01 deg
A >> P	220 mm
R >> L	220 mm
F >> H	149 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	6120.0 ms
Concatenations	1

Perf

Measurements	1
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Sequence - Part 1

Sequence Name	epfid
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Sequence - Part 1

Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1502 Hz/Px
Echo Spacing	0.77 ms
Free Echo Spacing	On
EPI Factor	128

Sequence - Part 2

Introduction	Off
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Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\t1_space
_sag_cs4_iso

TA: 2:49 min Coil Selection: Auto Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4.0 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	700.0 ms
TE	11.00 ms
Averages	1.0
Concatenations	1
AutoAlign	Head > Basis

Contrast - Common

TR	700.0 ms
TE	11.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	T1 Var
Fat-Water Contrast	Standard
Dark Blood	Off
Blood Suppression	Off
Wrap-up Magn.	Restore
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	256
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.0
Reference Scans	Integrated
Reference Lines PE	32
Reference Lines 3D	32
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	On
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	176
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	700.0 ms
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Basis

Geometry - AutoAlign

Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	250 mm
F >> H	250 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Image Scaling	1.000
Gain	High

Physio - Signal

1st Signal/Mode	None
Trigger Delay	0 ms
TR	700.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	spcR
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	630 Hz/Px
Echo Spacing	3.72 ms
Turbo Factor	42
Echo Train Duration	167 ms

Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	30 s

\\Research\Investigators - validated on FIT\Frederick\High bandwidth Siemens product scans\B1Map_f or_T1_mapping

TA: 11 sec Coil Selection: Auto Voxel Size: 3.1×3.1×8.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	20
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	5310.0 ms
TE	1.83 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	5310.0 ms
TE	1.83 ms
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm

Resolution - Common

Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	20
Distance Factor	50 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	8.0 mm
TR	5310.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

Geometry - Tim Planning Suite

Inline Composing	Off
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System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	tfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Low SAR
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	490 Hz/Px
Echo Spacing	4.12 ms
Asymmetric Echo	Allowed
Turbo Factor	64

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\T1Map_a anatomical

TA: 4:02 min Coil Selection: Auto Voxel Size: 0.2×0.2×3.0 mm³ Acc.: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
TE	2.70 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	15.00 ms
TE	2.70 ms
Flip Angle 1	5 deg
Flip Angle 2	26 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	2
Pause after Meas. 1	0.0 s
Multiple Series	Each Measurement

Contrast - Dynamic

3D Reordering	Standard
Time to Center	48.9 s
Burn Time to Center	Off

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	81 %
Trajectory	Cartesian
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	6/8
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter

Geometry - AutoAlign

Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
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System - Tx/Rx

? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	2
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	2
Pause after Meas. 1	0.0 s

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T1 Map
Auto Angle Calculation	On
T1 Estimate	800 ms
Flip Angle 1	5 deg
Flip Angle 2	26 deg
Measurements	2
Contrasts	1
TE	2.70 ms
TR	15.00 ms
Noise Threshold	15
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	280 Hz/Px
Asymmetric Echo	Weak
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\T1Map_p
2_anatomical

TA: 2:09 min Coil Selection: Auto Voxel Size: 0.2x0.2x3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Contrast - Dynamic

3D Reordering	Standard
Time to Center	26.0 s
Burn Time to Center	Off

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Slice Resolution	81 %
Trajectory	Cartesian
Interpolation	On

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
TE	2.71 ms
Averages	1
Concatenations	1
AutoAlign	---

Resolution - Acceleration

Acceleration Mode	GRAPPA
Total Factor	2
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	6/8
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Contrast - Common

TR	15.00 ms
TE	2.71 ms
Flip Angle 1	5 deg
Flip Angle 2	26 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	22
Phase Oversampling	0 %
Slice Oversampling	45.5 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	15.00 ms
Series	Ascending

Contrast - Dynamic

Dynamic Mode	Standard
Temporal Interpolation	1
Measurements	2
Pause after Meas. 1	0.0 s
Multiple Series	Each Measurement

Geometry - Common

Concatenations	1
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Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	2
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	2
Pause after Meas. 1	0.0 s

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T1 Map
Auto Angle Calculation	On
T1 Estimate	800 ms
Flip Angle 1	5 deg
Flip Angle 2	26 deg
Measurements	2
Contrasts	1

Inline - MapIt

TE	2.71 ms
TR	15.00 ms
Noise Threshold	15
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	280 Hz/Px
Asymmetric Echo	Weak
Optimization	Min. TE

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Incr. Gradient Spoiling	Off
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE

\\Research\Investigators - validated on FIT\Frederick\High bandwidth Siemens product scans\T2Map_a anatomical

TA: 6:35 min Coil Selection: Auto Voxel Size: 0.4×0.4×3.0 mm³ Acc.: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Contrast - Common

Reconstruction	Magnitude
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Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Routine

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	1910.0 ms
TE 1	13.80 ms
TE 2	27.60 ms
TE 3	41.40 ms
TE 4	55.20 ms
TE 5	69.00 ms
Averages	1
Concatenations	1
AutoAlign	---

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	4/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	25
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	1910.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Contrast - Common

TR	1910.0 ms
TE 1	13.80 ms
TE 2	27.60 ms
TE 3	41.40 ms
TE 4	55.20 ms
TE 5	69.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	180 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	5

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1910.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T2 Map
Flip Angle	180 deg
Measurements	1
Contrasts	5
TE 1	13.80 ms
TE 2	27.60 ms
TE 3	41.40 ms
TE 4	55.20 ms
TE 5	69.00 ms
TR	1910.0 ms
Noise Threshold	15
Save Original Images	On

Sequence - Part 1

Sequence Name	se
RF Pulse Type	Low SAR
Gradient Mode	Fast

Sequence - Part 1

Bandwidth	228 Hz/Px
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Sequence - Part 2

Introduction	On
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Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\T2StarMa
p_anatomical

TA: 2:52 min Coil Selection: Auto Voxel Size: 0.4×0.4×3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	11
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	445.0 ms
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	445.0 ms
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
MTC	Off
Magn. Preparation	None
Flip Angle	60 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	5

Contrast - Common

SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	11
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	445.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter

Geometry - AutoAlign

Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	445.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T2* Map
Flip Angle	60 deg
Measurements	1
Contrasts	5
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
TR	445.0 ms
Noise Threshold	25
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Readout Mode	Monopolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Bandwidth 1	260 Hz/Px
Bandwidth 2	260 Hz/Px
Bandwidth 3	260 Hz/Px
Bandwidth 4	260 Hz/Px
Bandwidth 5	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\High bandwidth Siemens product scans\\T2StarMa
p_p2_anatomical

TA: 1:32 min Coil Selection: Auto Voxel Size: 0.4x0.4x3.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	11
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	445.0 ms
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
Averages	1
Concatenations	1
AutoAlign	---

Contrast - Common

TR	445.0 ms
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
MTC	Off
Magn. Preparation	None
Flip Angle	60 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	5

Contrast - Common

SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	11
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	160 mm
FOV Phase	100.0 %
Slice Thickness	3.0 mm
TR	445.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS Restricted
Radial Sorting	Off
MSMA	S - C - T
Sagittal	Med >> Lat
Coronal	A >> P
Transversal	H >> F
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - pTx

LR Balancing	Off
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	445.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	160 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off

Inline - Soft Tissue

Measurements	1
--------------	---

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	T2* Map
Flip Angle	60 deg
Measurements	1
Contrasts	5
TE 1	4.36 ms
TE 2	11.90 ms
TE 3	19.44 ms
TE 4	26.98 ms
TE 5	34.52 ms
TR	445.0 ms
Noise Threshold	25
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Readout Mode	Monopolar
Gradient Mode	Fast
Flow Compensation 1	None
Flow Compensation 2	None
Flow Compensation 3	None
Flow Compensation 4	None
Flow Compensation 5	None
Bandwidth 1	260 Hz/Px
Bandwidth 2	260 Hz/Px
Bandwidth 3	260 Hz/Px
Bandwidth 4	260 Hz/Px
Bandwidth 5	260 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s